

Exercises for Section 10.1

Directions: For exercises 1 - 6, identify the center, vertices, and foci of the given ellipses.

1. $\frac{x^2}{4} + \frac{y^2}{49} = 1$

2. $\frac{x^2}{100} + \frac{y^2}{64} = 1$

3. $\frac{(x-2)^2}{49} + \frac{(y-4)^2}{25} = 1$

4. $\frac{(x-2)^2}{81} + \frac{(y+1)^2}{16} = 1$

5. $\frac{(x+5)^2}{4} + \frac{(y-7)^2}{9} = 1$

6. $\frac{(x-7)^2}{49} + \frac{(y-7)^2}{49} = 1$

Directions: For exercises 7 - 11, graph the given ellipses, noting center, vertices, and foci.

7. $\frac{x^2}{25} + \frac{y^2}{36} = 1$

8. $\frac{x^2}{16} + \frac{y^2}{9} = 1$

9. $\frac{(x-2)^2}{64} + \frac{(y-4)^2}{16} = 1$

10. $\frac{(x+3)^2}{9} + \frac{(y-3)^2}{9} = 1$

11. $\frac{x^2}{2} + \frac{(y+1)^2}{5} = 1$

Directions: For exercises 12 - 13, use the given information about the graph of each ellipse to determine its equation.

12. Center at the origin, symmetric with respect to the x - and y -axes, focus at $(4,0)$, and point on graph $(0,3)$.

13. Center at the origin, symmetric with respect to the x - and y -axes, focus at $(0,-2)$, and point on graph $(5,0)$.